EMERGENCY PROCEDURES & WILDERNESS MEDICAL

Breakdown, fire, medical, snakebite & environmental emergencies

2009 LC76 SW · 1HZ 4.2L Diesel · low-boost turbo (0.6 bar) · intercooler
Last updated: 12 October 2026

STOP STOP · THINK · OBSERVE · PLAN **C-ABC** BLEEDING · AIRWAY · BREATHING · CIRCULATION **EVACUATE?** ASSESS · DECIDE · ACT

☎ EMERGENCY NUMBERS — READ FIRST

WHICH LIST TO USE

The number that connects depends on the phone in your hand, not on where you are standing

Two lists follow. **In South Africa, with cell coverage**, use the cellphone list. **On the satellite phone** — anywhere, including inside SA where there is no cell signal — use the satphone list. They are not interchangeable: a national share-call code (0860 / 0861 / 082 / 084 / 112) **cannot be routed by the satellite phone** and will simply fail.

These numbers are a controlled copy. **R29 §6 is the original** — if anything here disagrees with it, R29 wins. The full evacuation drill, the handover script and the four-layer medical model are in §9 below and R29 §6–§8.

📱 IN SOUTH AFRICA — FROM A CELLPHONE

Works on any SA mobile network with coverage

CALL	NUMBER	FOR
Universal emergency	112	Any emergency, any SA network
ER24	084 124	Medical / ambulance
Genesis — Neil	0861 436 374	Neil's medical aid (ER24-operated)
Bonitas SOS — Mary-Ann	0860 555 505	Mary-Ann's medical aid (AZOZA ambulance)
Cross Country (vehicle & assistance)	+27 11 966 5004 (satphone) · 0800 005 688 (in-country only)	Roadside, tow, cross-border repatriation, medical assist/evac. Call the centre FIRST. Medevac has a 30-day / over-75 limit abroad — R29 §6.

Membership / policy numbers — fill in on the printed card: Genesis ___ · Bonitas ___ · Cross Country CO16591

🛰 ON THE SATELLITE PHONE — IN SA OR ABROAD

The Karoo case: inside SA and outside cell coverage at the same time

None of the numbers above will connect from the satphone. Not 112, not the 082 / 084 codes, not any 0860 or 0861 number — a satellite handset cannot route a national share-call code. Dial these full numbers instead:

CALL	NUMBER	FOR
ER24 dispatch	+27 10 205 3000	Medevac coordination · Neil's Genesis route abroad
AMREF Flying Doctors	+254 730 811 811 · +254 709 962 811	Medevac — East Africa
Bonitas / AZOZA — Mary-Ann	+27 11 991 8174	AZOZA international travel desk
Cross Country	+27 11 966 5004	satphone-dialable JHB landline; 0800 005 688 is in-country only

Numbers owned by R29 §6. Provider lines are provider-published; the medevac test-dial regime is RA-10.

1. EMERGENCY PRIORITIES FRAMEWORK

🧭 MEDICAL QUICK-INDEX — WHERE TO LOOK FIRST

Find the right protocol fast. For the triage sequence and the evacuate-now list, use the STOP / ABC / Evacuation Decision cards below.

Road accident, rollover, extrication, pelvic binding → §4 Road Accident & Rollover (this guide)
Snake bite, scorpion, or venom spat in the eyes → §6 Snake Bite Protocols (this guide)
Trauma, severe bleeding, burns, fractures, CPR, choking, anaphylaxis, shock, recovery position → §5 Medical Emergencies (this guide)
Heat, cold/hypothermia, altitude, drowning, lightning → §7 Environmental Emergencies (this guide)
Fever, malaria, diarrhoea, bilharzia, tick-bite, dengue → R17 Health & Hygiene
Unsure whether to self-treat or reach a clinic → R17 "When to Seek Medical Help"
Trauma / first-aid kit → §10 (this guide) · **full medication & health kit** → R17 Medical Kit
Evacuation & casevac logistics, insurance, AMREF → §9 Communication & Evacuation (this guide) · R29 Insurance & Medevac

🛑 STOP FRAMEWORK

Before any action in an emergency, follow this sequence

S — STOP
Stop what you're doing. Do not panic. Take a breath.

T — THINK
What has happened? What are the immediate dangers? What resources do I have?

O — OBSERVE
Look around. Assess the scene. Check for hazards. Count all persons.

P — PLAN
Make a plan before acting. Prioritize actions. Communicate the plan.

C-ABC PRIORITIES

Catastrophic bleeding first, then airway — for any injured person, in this order

⟨C⟩ — CATASTROPHIC BLEEDING

- Spurting or pooling blood? Control it BEFORE anything else
- Direct pressure → wound packing → tourniquet (§5 Severe Bleeding)
- A casualty can bleed out in the time it takes to assess breathing

A — AIRWAY

- Is the airway clear and open?
- If unconscious: head tilt, chin lift
- Clear any obstructions (vomit, blood, debris)

B — BREATHING

- Look, listen, feel for breathing
- If not breathing: begin rescue breaths
- Count respiratory rate (normal: 12–20/min)

C — CIRCULATION

- Check for pulse (carotid or radial)
- Look for severe bleeding—control immediately
- Check skin colour, temperature, capillary refill

EVACUATION DECISION

When to evacuate vs treat in field

EVACUATE IMMEDIATELY for:

- Severe bleeding that won't stop
- Difficulty breathing / airway compromise
- Chest pain or signs of heart attack
- Stroke symptoms (face droop, arm weakness, speech)
- Severe allergic reaction (anaphylaxis)
- Suspected spinal injury
- Altered consciousness / severe head injury
- Snake bite from neurotoxic species
- Signs of appendicitis

CAN TREAT IN FIELD:

- Minor cuts and abrasions
- Mild dehydration (if water available)
- Minor sprains and strains
- Mild heat exhaustion (if shade/water available)
- Insect stings (no allergic reaction)

2. VEHICLE BREAKDOWN PROCEDURES

IMMEDIATE ACTIONS

First steps when the vehicle stops

1. Move vehicle off road if possible (coast if engine dead)
2. Activate hazard lights immediately
3. Apply handbrake, place in gear (manual) or Park
4. Exit vehicle safely—away from traffic
5. Place warning triangle 50m+ behind vehicle
6. Assess the situation: smoke, fluids, damage?
7. Do not open bonnet if smoke is present (fire risk)

INITIAL ASSESSMENT

What to check and document

Check for:

- Fluid leaks (fuel, oil, coolant, brake fluid)
- Smoke or unusual smells
- Warning lights that were on before failure
- Any unusual sounds (knocking, grinding, hissing)
- Damage from impact or terrain

Document:

- Exact GPS coordinates
- Time of breakdown
- Symptoms before failure
- Photos of any visible damage

DECISION: REPAIR VS WAIT

When to attempt field repair

Attempt field repair if:

- Problem is simple and you have parts/tools
- You have relevant knowledge/experience
- Conditions are safe (weather, security, light)
- Repair won't make things worse

Wait for help if:

- Major mechanical failure (engine, gearbox)
- You lack knowledge or tools
- Conditions are dangerous
- Repair could worsen the situation
- Help is accessible and within reasonable time

🚗 STAY WITH VEHICLE VS WALK OUT

Critical decision that can mean life or death

STAY WITH VEHICLE when:

- You are not in immediate danger
- People know where you are
- Vehicle provides shelter from elements
- You have water and supplies
- Distance to help is unknown or far
- Terrain is difficult or dangerous
- You are injured or unfit for walking

A vehicle is easier to spot from the air than a person. Search parties will look for your vehicle first.

WALK OUT only when:

- Vehicle is in immediate danger (fire, flood)
- No one knows your location
- You have no means of signalling
- You know exactly where help is (visible)
- You have sufficient water and supplies
- You are physically capable
- Weather conditions are favorable

Walking out is almost always the wrong choice. People have died walking for help when rescuers found their vehicle within hours.

3. VEHICLE FIRES

Ownership: this section owns the in-the-moment drill. Prevention, extinguisher sizing & service, fire blanket & aerosol suppressors, fuel & gas handling and camp-fire discipline are owned by **R42 Fire Safety Guide** — read it before you need this page.

🔥 ENGINE FIRE

Most common vehicle fire

1. Pull over immediately, turn off engine
2. Get all occupants out and away (50m minimum)
3. Isolate the batteries — **M-series battery switch OFF** (the one-second action; kills the house bank, MultiPlus included). **No crank isolator is fitted** — spanner on the crank negative only if the terminal is safely reachable (R42 §2)
4. DO NOT open bonnet—oxygen feeds fire
5. If small fire visible: discharge extinguisher through grille gaps
6. If fire is large or spreading: evacuate further, call for help
7. Never attempt to fight a fuel-fed fire
8. Stay upwind of smoke and fumes

WARNING: Vehicle fires can cause fuel tank explosion. Move all persons at least 50m away if fire cannot be immediately extinguished.

🌡️ TURBO/EXHAUST FIRE RISK (1HZ SAC TURBO)

Your aftermarket turbo runs hot — exhaust manifold can reach 600°C under load

Fire risk factors for turbocharged 1HZ:

- Exhaust manifold and turbo housing glow red-hot under sustained load
- Oil or fuel leaks near turbo = immediate fire risk
- Heat shields must be intact — check regularly
- Grass/debris under vehicle can ignite from exhaust heat

Prevention — check these regularly:

- No oil leaks around turbo oil feed/drain lines
- Exhaust manifold heat shields in place and tight
- No cracked or loose exhaust connections
- Monitor EGT gauge — back off at ≥550°C sustained; 650°C pre-turbo is the hard ceiling (thresholds: R21)
- Clear grass and debris from undercarriage

If you smell hot oil or see smoke from engine bay:

1. Pull over safely, away from dry grass
2. Switch off engine immediately
3. Exit and move away — turbo is extremely hot
4. DO NOT open bonnet immediately — let turbo cool
5. After 5+ minutes, check for leaks or burning

> **Note:** A red-glowing turbo at night is normal under heavy load (towing, steep climbs). However, smoke or flame is NOT normal — stop immediately.

⛽ FUEL FIRE

Extremely dangerous—do not fight

1. Evacuate all persons immediately
2. Move at least 100m away
3. DO NOT attempt to extinguish
4. Call emergency services if available
5. Warn any approaching vehicles
6. Stay upwind of smoke

Fuel fire characteristics:

- Burns extremely hot
- Spreads rapidly
- Can reignite after appearing extinguished
- Fuel vapours are explosive

⚡ ELECTRICAL FIRE

Smouldering wires, burning plastic smell

1. Turn off ignition immediately
2. Isolate the batteries — **M-series battery switch OFF** (one-second action, kills the house bank). No crank isolator is fitted: spanner on the crank negative terminal ONLY if safely reachable (R42 §2)
3. Exit vehicle with all occupants
4. Use dry powder extinguisher on wiring/plastic fires
5. If fire spreads: evacuate and wait

> **LiFePO4 house battery involved (210 Ah pack):** dry powder knocks down surface flame only — a venting cell is self-oxidising and will re-flash. **WATER IN VOLUME IS CORRECT on LiFePO4** — it cools the cells and is the only thing that stops thermal runaway. If the pack itself is burning, expect re-flash for hours: evacuate, treat as a total-loss fire, and manage the perimeter (R42 §2).

Signs of electrical fire:

- Burning plastic smell
- Melting wire insulation
- Dashboard warning lights flickering
- Smoke from dashboard or under seats

🔥 BUSH/GRASS FIRE ESCAPE

Wildfire can overtake a vehicle—know your escape options

If fire is approaching:

1. Turn on headlights and hazard lights
2. Close all windows and vents
3. Turn off air conditioning (draws smoke in)
4. If road is clear: drive through to burned area (safer)
5. Drive slowly—visibility may be near zero

If trapped by fire:

1. Park in area with least vegetation
2. Turn off engine but leave ignition on (for windows)
3. Close all windows and vents tightly
4. Get down below window level
5. Cover with wool blanket if available

6. Wait until fire passes—vehicle provides protection

Vehicle survival: A vehicle will provide protection from radiant heat as fire passes over. The fuel tank will not explode from external heat. Stay inside until the fire front passes, then exit carefully (metal will be hot).

4. ROAD ACCIDENT & ROLLOVER

🚨 THE STATISTICAL REALITY

The vehicle — not snakes, hippos or heat — is the most likely cause of serious injury or death on an overland trip

Road traffic accidents kill more overlanders than every wildlife and environmental hazard combined. High-speed gravel, wandering livestock, unlit trucks, night driving fatigue and rollover on soft verges are the dominant mechanisms. Everything in this section assumes what the rest of this guide assumes: help is 4–12+ hours away and the two of you are the first responders — possibly both injured.

🚗 IMMEDIATE ACTIONS — YOUR OWN CRASH

First 5 minutes: scene safety, fire, then people

1. STOP framework (§1) — do not act before you think
2. Switch off ignition; isolate the battery switch if reachable
3. Check for fuel smell, smoke, leaking diesel — fire risk decides everything
4. Account for both persons — speak to each other, injuries stated out loud
5. If on a live road: hazards on, triangle 50m+ back, stay off the carriageway
6. Only then begin casualty assessment — C-ABC (§1)

BOTH INJURED: The less-injured person treats catastrophic bleeding first (own or partner's), then activates the InReach SOS BEFORE further treatment. Coordination started early is worth more than ten minutes of perfect first aid.

💥 BLOW-OUT AT SPEED

Stop it becoming a rollover

1. No hard braking — feet off the pedals
2. Hold the wheel straight; keep off the camber
3. Change down one gear gently as the vehicle slows; stop well off the road

Hard braking in a blow-out leads to a slide, and a slide into even a shallow rut can roll the vehicle. Full doctrine: R28 §5 (St Pierre White 2014, p. 197). Wheel change: R40.

🩹 MOVE THE CASUALTY — OR NOT?

The extrication decision in one rule

MOVE IMMEDIATELY (spinal risk accepted) if:

- Fire, smoke, or strong fuel smell
- Vehicle unstable and may roll/slide further
- Rising water
- Airway cannot be managed in place (unconscious, face-down, slumped)

Otherwise DO NOT move:

- Support the head/neck in the position found
- Treat bleeding and monitor breathing in place
- Wait for coordinated evacuation

Rule: A live casualty with a possible spinal injury beats a perfectly immobilised casualty in a burning vehicle. Threat to life overrides spinal caution — nothing else does.

🦴 PELVIC INJURY — THE SILENT BLEEDER

High-energy impact or rollover + hip/groin pain = treat as broken pelvis

Suspect a pelvic fracture after any high-energy crash with: pain in hips, groin or lower back; inability to stand; legs rotated outward. An unstable pelvis can bleed 2+ litres internally over the hours you are waiting — it is survivable IF bound early.

Improvised pelvic binder:

1. Do NOT "spring" or rock the pelvis to test it — assess by mechanism and pain only
2. Slide the binder material under the natural hollows (knees, lower back) before positioning — minimise log-rolling
3. Wrap a wide band (ratchet strap with cloth padding, rolled towel + tape, SAM sling) around the pelvis at the level of the *greater trochanters* — the bony points at the top of the outer thighs, NOT the waist
4. Tighten firmly (snug, not crushing), tie or clip in front
5. Bind knees and ankles together with padding between
6. Keep flat, keep still, keep warm — evacuate urgently

🔄 ROLLOVER — EXITING AN INVERTED VEHICLE

Orientation, fuel and glass are the hazards once motion stops

1. Wait for all movement to stop; engine off if still running
2. Check yourself before releasing the belt — moving may reveal injuries
3. Brace one hand on the roof/headliner, release seatbelt in a controlled drop
4. Exit through the easiest opening — kick out a windscreen corner or rear glass if doors are jammed
5. Get clear: fuel and battery acid run where they should not in an inverted vehicle
6. Do NOT attempt to right or re-enter the vehicle until fuel/fire risk is assessed

Note: Diesel is far less volatile than petrol — a fuel smell after a rollover is a warning, not an automatic fireball. Assess calmly, but assess.

👥 THIRD-PARTY ACCIDENTS & SCENE SAFETY

Helping at someone else's crash — and when not to stop

If you stop to help:

- Park protectively (your vehicle shields the scene), hazards on
- Triage by noise: the quiet casualty is usually the sickest — check the silent ones first
- Catastrophic bleeding → airway → breathing, casualty by casualty
- Delegate: bystanders hold pressure, control traffic, note details

Scene judgement:

- If a crowd is hostile or the scene feels unsafe, do not stop — drive to the next police post or town and report with GPS coordinates
- If you are involved in an accident and the scene turns hostile, the same applies: drive to the nearest police station and report there (see §11 Security Incidents)

5. MEDICAL EMERGENCIES

🩸 SEVERE BLEEDING

Life-threatening blood loss requires immediate action

1. Apply direct pressure with clean cloth/bandage
2. Maintain firm, constant pressure—do not remove to check
3. If blood soaks through: add more dressings on top
4. If pressure fails: PACK the wound (below), then pressure over the packing
5. If packing + pressure fails on a limb: apply tourniquet

Tourniquet use (life-threatening limb bleeding):

- Apply 5–8 cm above wound (not on joint)
- Tighten until bleeding stops
- Note time of application
- Do NOT remove once applied
- Evacuate immediately

Wound packing (junctional wounds — groin, armpit, neck — where a tourniquet cannot go):

- Push haemostatic gauze (§10 kit) — or plain gauze — DEEP into the wound, onto the bleeding point
- Keep feeding gauze in until the cavity is full; it will feel wrong — do it anyway
- Firm pressure on the packed wound for 3+ minutes, then pressure dressing over it
- Do not remove packing in the field — it goes to hospital with them

🦴 FRACTURES

Immobilize and evacuate

Signs of fracture:

- Pain and tenderness at site
- Swelling and bruising
- Deformity or unnatural angle
- Inability to move or bear weight
- Grinding sensation (crepitus)

Treatment:

- Immobilize the limb in position found — EXCEPT: if the limb beyond a deformed fracture is pulseless, white or blue, make ONE gentle attempt to realign with steady pull along the limb's axis, then splint; recheck pulse
- Splint above and below the break
- Use SAM splint, sticks, or padding
- Check circulation beyond splint (pulse, colour, sensation)
- Pain relief if available (paracetamol + ibuprofen together; stronger prescription analgesia per R17 kit)
- Evacuate for X-ray and proper treatment
- OPEN fracture (bone through skin): rinse gross dirt with clean water, cover with moist sterile dressing, splint, start the R17 antibiotic — do NOT push bone back in

🧠 HEAD INJURY

Concussion and worse require evacuation

Signs requiring evacuation:

- Loss of consciousness (any duration)
- Confusion or memory loss
- Unequal pupil size
- Clear fluid from nose or ears
- Worsening headache
- Repeated vomiting
- Seizures
- Weakness on one side of body

Care while awaiting evacuation:

- Keep person still and calm
- Assume spinal injury—immobilize neck
- Monitor consciousness level frequently
- Do not give food or drink

🔄 RECOVERY POSITION

Unconscious but BREATHING — this position keeps them alive while you wait

Any unconscious, breathing casualty left on their back can die of a blocked airway or inhaled vomit during a multi-hour wait. This is the single highest-value, lowest-skill intervention in this guide.

1. Kneel beside them; straighten the legs; nearest arm out at a right angle, palm up
2. Far arm across the chest, back of that hand against their near cheek — hold it there
3. Grip the far knee, pull it up, and roll them toward you onto their side
4. Tilt the head back slightly to keep the airway open; top knee bent as a prop
5. Check breathing every few minutes; swap sides every ~30 minutes on a long wait

> **Suspected spinal injury:** stay in position found with the head supported IF they are breathing well. If the airway is at risk (vomiting, gurgling) the airway WINS — roll them as one unit, keeping head and spine in line.

🔥 BURNS

Cool, cover, and transport

1. Remove from heat source / stop the burning
2. Cool burn with clean, cool water for 20 minutes
3. Remove clothing/jewelry near burn (unless stuck)
4. Cover with clean, non-stick dressing
5. Do NOT apply ice, butter, or creams
6. Do NOT burst blisters

Evacuate for:

- Burns larger than palm size
- Burns to face, hands, feet, genitals, joints
- Burns encircling a limb
- Chemical or electrical burns
- Burns with smoke inhalation

😰 SHOCK

Life-threatening—recognize and treat

Signs of shock:

- Pale, cold, clammy skin
- Rapid, weak pulse
- Rapid, shallow breathing
- Confusion, anxiety, restlessness
- Thirst
- Nausea or vomiting

Treatment:

- Lay person down, elevate legs (if no spinal injury)
- Control any external bleeding
- Keep warm with blanket—prevent heat loss
- Reassure and keep calm
- Do NOT give food or drink
- Evacuate urgently

🦠 SUSPECTED APPENDICITIS

Surgical emergency—evacuate immediately

Signs and symptoms:

- Pain starting around navel, moving to lower right
- Pain worsens with movement, coughing
- Rebound tenderness (pain when pressure released)
- Loss of appetite, nausea, vomiting
- Low-grade fever
- Inability to pass gas

URGENT: If appendicitis is suspected, evacuate immediately. A ruptured appendix is life-threatening. Do NOT give food or water. Keep person as still as possible. If evacuation will exceed ~6 hours, start the R17 broad-spectrum antibiotic with a sip of water — it buys time, it does not replace surgery.

❤️ CPR – CARDIAC ARREST

Unresponsive and not breathing normally — start at once

Confirm arrest:

- Unresponsive AND not breathing normally — occasional gasps are NOT normal breathing
- Shout for help and trigger emergency comms (§9); grab the kit and CPR face shield (§10)

Compressions and breaths — 30:2:

- Heel of one hand on the centre of the chest (lower half of the breastbone), other hand on top
- Push hard and fast — depth 5–6 cm, rate 100–120 per minute, let the chest recoil fully each time
- 30 compressions, then 2 rescue breaths (tilt head, lift chin, pinch nose, seal, ~1 second each)
- Continue 30:2 without stopping; swap rescuer every ~2 minutes if you can, to keep compressions strong
- Untrained or unwilling to give breaths in a cardiac (non-drowning) arrest: give continuous compressions only

REALITY: Without a defibrillator and advanced care, field survival from cardiac arrest is low. Start CPR and get emergency comms running (§9) — a lone rescuer activates the InReach SOS within the FIRST two minutes, not after exhaustion. Continue until help arrives, the person recovers, or you cannot go on. A lone rescuer who has given 30 minutes of good CPR in a normal-temperature cardiac arrest, with no response and no advanced help coming, may stop — that is the accepted wilderness standard, and it is not giving up. EXCEPTIONS — continue much longer for: hypothermia (§7), drowning (§7 protocol — breaths first), and lightning strike (§7).

😵 CHOKING – AIRWAY OBSTRUCTION

Sudden silence and clutching the throat is the classic sign

If they can cough, speak, or breathe (mild):

- Encourage them to keep coughing — do not interfere

If they cannot cough, speak, or breathe (severe) and are conscious:

- 5 sharp back blows between the shoulder blades with the heel of your hand
- If that fails: 5 abdominal thrusts — hands just above the navel, sharp inward-and-upward pulls
- Alternate 5 back blows and 5 abdominal thrusts until the object clears
- Pregnant or very large casualty: use chest thrusts instead of abdominal thrusts

IF THEY GO UNCONSCIOUS: Lower them to the ground and start CPR (above) at once — compressions may also dislodge the blockage. Trigger emergency comms (§9).

💉 ANAPHYLAXIS – SEVERE ALLERGIC REACTION

Rapid onset after a sting, food, or drug — adrenaline saves lives

Recognise (rapid onset, any of):

- Airway: throat or tongue swelling, tight throat, hoarse voice, noisy breathing
- Breathing: wheeze, sudden breathlessness
- Circulation: pale, faint, dizzy, collapse
- Often with skin signs (hives, flushing, swelling) — but skin signs can be absent

Treat — do not delay:

- Inject adrenaline into the mid-outer thigh at once: EpiPen (§10), or 0.5 mg of 1:1000 (0.5 mL) by syringe
- Lie the person flat and raise their legs; if breathing is hard let them sit up — **never stand or sit them up suddenly**
- Repeat the adrenaline after 5 minutes if there is no improvement
- Antihistamines help the skin only — they are NOT a substitute for adrenaline

EVACUATE URGENTLY: A second (biphasic) reaction can follow hours later. Get emergency comms running (§9) and keep the used auto-injector. Standing a collapsing casualty upright can be fatal.

6. SNAKE BITE PROTOCOLS

🐍 CRITICAL: SNAKEBITE FIRST AID

Southern Africa has both neurotoxic and cytotoxic snakes—treatment differs

DO NOT:

- Cut the bite or try to suck out venom
- Apply a tourniquet (except as pressure bandage for mambas/cobras)
- Apply ice or electric shock
- Give alcohol
- Try to catch or kill the snake (but photograph if safe)

NEUROTOXIC SNAKES (Black Mamba, Cape Cobra)

⚠️ USE PRESSURE BANDAGE

- Apply pressure bandage immediately
- Wrap firmly from bite site up the limb
- Pressure should be like a firm ankle wrap
- Splint the limb to prevent movement
- Do NOT remove until at hospital
- Evacuate URGENTLY—death can occur in 30 min to 8 hrs
- Be prepared to support breathing

CYTOTOXIC SNAKES (Puff Adder, Spitting Cobra)

⚠️ NO PRESSURE BANDAGE

- Keep victim calm and still
- Remove rings/jewelry near bite (swelling)
- Keep the limb at heart level — neither raised nor hanging
- Do NOT apply pressure bandage
- Clean wound gently
- Transport to hospital
- Antivenom needed for severe cases

🐍 SOUTHERN AFRICAN DANGEROUS SNAKES

Know the major species and their venom type

SNAKE	VENOM TYPE	DANGER LEVEL	PRESSURE BANDAGE?	NOTES
Black Mamba	Neurotoxic	**EXTREME**	YES	Fastest snake. Death in 30 min–8 hrs. Most deaths.
Cape Cobra	Neurotoxic	**EXTREME**	YES	As potent as mamba. Rears with hood.
Green Mamba	Neurotoxic (predominant)	**VERY HIGH**	YES	Tree-dwelling. Shy but dangerous.
Puff Adder	Cytotoxic	HIGH	**NO**	Most bites in SA. Lies still, stepped on. Severe tissue damage.
Mozambique Spitting Cobra	Cytotoxic	HIGH	**NO**	Most serious bites in SA. Can spit venom at eyes.
Boomslang	Haemotoxic	**VERY HIGH**	Disputed	Causes bleeding. Separate antivenom required. Shy, rarely bites.
Gaboon Adder	Cytotoxic	HIGH	**NO**	Massive fangs. Found in forests. Docile but dangerous.
Night Adder	Cytotoxic (mild)	MODERATE	**NO**	Common. Painful but rarely life-threatening.

Important: About 12 snakebite deaths per year in Southern Africa—mostly from Black Mamba and Cape Cobra. Most bites do NOT result in full envenomation. Stay calm and get to hospital.

👁️ SPITTING COBRA - VENOM IN EYES

Mozambique Spitting Cobra and Rinkhals

1. Rinse eye immediately with large amounts of water
2. Continue rinsing for 15–20 minutes
3. Any bland liquid works: water, milk, saline
4. Hold eyelids open—victim will resist
5. Apply antibiotic eye drops if available
6. Cover with eye pad
7. Transport to hospital for examination
8. Same irrigation approach for any eye foreign body (thorn, grit) — never rub; if embedded, cover both eyes and evacuate

Recovery: With prompt irrigation, most people regain full sight within a few days. Without treatment, corneal damage can cause permanent blindness.

🦂 SCORPION STINGS

Painful but rarely life-threatening in Southern Africa

Treatment:

- Clean the sting site
- Apply cold pack for pain relief
- Pain relief medication (paracetamol, ibuprofen)
- Monitor for allergic reaction
- Most stings resolve in 24–48 hours

Seek medical help if:

- Difficulty breathing
- Muscle spasms or twitching
- Drooling or difficulty swallowing
- Sting in child under 10 years

7. ENVIRONMENTAL EMERGENCIES

🌡️ HEAT STROKE VS HEAT EXHAUSTION

Critical distinction—heat stroke kills

FEATURE	HEAT EXHAUSTION	HEAT STROKE
Skin	Cool, pale, sweating	**Hot, red, DRY**
Temperature	<40°C	**>40°C**
Mental state	Alert but fatigued	**Confused, agitated, unconscious**
Sweating	Present	**Often absent**
Severity	Serious	**LIFE-THREATENING**

Heat Exhaustion Treatment:

- Move to shade
- Lie down, elevate legs
- Remove excess clothing
- Cool with water, fan
- Give water with electrolytes
- Recovery usually within 30 minutes

Heat Stroke Treatment (EMERGENCY):

- Call for evacuation immediately
- Move to shade
- Cool aggressively: immerse in water, wet sheets, fan
- Pack ice at neck, armpits, groin
- Do NOT give fluids if unconscious
- This is a medical emergency—evacuate NOW

💧 DEHYDRATION

Common and dangerous in African heat

Signs of dehydration:

- Thirst (late sign—don't wait for this)
- Dark urine, reduced urine output
- Dry mouth and lips
- Headache, dizziness
- Fatigue, weakness
- Rapid heartbeat

Treatment:

- Oral rehydration solution (ORS) is ideal
- Or water with salt and sugar (1L water + ½ tsp salt + 6 tsp sugar)
- Sip slowly—don't gulp
- Rest in shade
- Monitor urine colour (should become lighter)

> **Prevention:** Drink before you're thirsty. 2–3 litres per day minimum in hot conditions. More if active. Urine should be pale yellow.

🦟 MALARIA RECOGNITION

Endemic in much of Southern/East Africa

Symptoms (appear 7–30 days after bite):

- Fever (often cyclic: chills → fever → sweating)
- Severe headache
- Muscle and joint pain
- Fatigue and weakness
- Nausea, vomiting, diarrhea

Action:

- If fever develops in malaria area: assume malaria until proven otherwise
- Begin treatment dose of antimalarials if available
- Get to medical facility for blood test
- Malaria can progress rapidly—do not delay

🐘 LARGE ANIMAL ENCOUNTERS

Hippo, elephant, buffalo—Africa's most dangerous

General rules:

- Never get between hippo and water
- Never approach elephant with ears out, trunk up
- Buffalo: most unpredictable—keep extreme distance
- Stay in vehicle—it's your protection
- Keep windows up if animal approaches
- Do NOT rev engine or make sudden movements

If charged:

- In vehicle: reverse slowly, don't turn your back
- On foot (worst case): find cover, climb if possible
- Elephant mock charges are common—stand still, back away slowly
- If contact: play dead (curl up, protect head)

🌊 DROWNING / NEAR-DROWNING

Breaths first — this is a hypoxic arrest, not a cardiac one

Get them out safely:

- Rescuer safety first — reach or throw, do not become the second casualty
- Remove from the water and keep them horizontal where possible

If not breathing normally:

- Open the airway and give **5 initial rescue breaths first** (gasping is not normal breathing)
- Then start CPR at 30 compressions : 2 breaths — depth 5–6 cm, rate 100–120/min
- Do **not** use compression-only CPR for drowning — the breaths are what they need
- Expect vomiting: turn the head and body to the side, clear the mouth, then continue
- Do not waste time trying to "push the water out" of the lungs

> **EVACUATE EVEN IF THEY RECOVER:** Anyone who needed rescuing or rescue breaths must reach medical care — delayed lung complications ("secondary drowning") can develop over the next 6–24 hours. Trigger emergency comms (§9).

🥶 HYPOTHERMIA

Karoo and highveld winter nights go well below zero — a wet, injured or immobile person cools fast

Recognise (progression):

- Mild: shivering, clumsy hands, withdrawn ("the umbles" — mumbles, stumbles, fumbles)
- Moderate: shivering STOPS, confusion, slurred speech, poor decisions
- Severe: unconscious, barely detectable pulse and breathing

Treatment:

1. Out of wind and wet — into the vehicle or tent
2. Replace wet clothing; insulate from the ground
3. Wrap: sleeping bag + space blanket over, hat on
4. Conscious and shivering: warm sweet drinks, food — shivering rewarms
5. Not shivering / confused: handle GENTLY (rough movement can stop a cold heart), warm packs to chest and armpits only, evacuate

> **COLD ARREST:** A severely hypothermic casualty who appears dead may not be. Continue CPR far beyond normal limits — "not dead until warm and dead." This is one of the stated exceptions to stopping CPR (§5).

🏔️ ALTITUDE ILLNESS

East African plateau routes run 1,500–2,500m+; Mt Kenya foothills higher

Acute mountain sickness (from ~2,500m): headache + nausea, poor sleep, fatigue within hours of arrival. Treat: stop ascending, rest, hydrate, paracetamol. Resolves in 1–2 days.

Danger signs — descend immediately:

- Breathlessness at rest, wet cough (fluid on lungs)
- Confusion, staggering walk, severe headache unrelieved by painkillers

> **The only reliable treatment is DESCENT.** With a vehicle this is minutes, not days — drive down 500–1,000m at the first danger sign. Vehicle-based travel makes altitude illness rare but not impossible after a fast climb to a high camp.

⚡ LIGHTNING

The highveld and Zambian plateau have among the highest strike densities on Earth

Avoidance — the 30/30 rule: if thunder follows flash within 30 seconds (≈10 km), take shelter; wait 30 minutes after the last thunder.

- The VEHICLE is excellent shelter (metal shell, not the tyres) — get in, windows up, don't touch metal
- Never shelter under isolated trees; avoid ridgelines and open ground
- Caught in the open: crouch on the balls of your feet, feet together, spread the group 15m+ apart
- Tents give NO lightning protection — move to the vehicle

REVERSE TRIAGE: Lightning inverts normal triage. A strike victim in cardiac arrest has an unusually GOOD chance — the strike stuns the heart and breathing centre, and CPR bridges the gap. Treat the "dead" FIRST. Casualties carry no charge — touch them freely. Start CPR at once (§5) and continue longer than you normally would.

8. WATER EMERGENCY PROCEDURES

💧 ENGINE HYDRO-LOCK PREVENTION

Water in engine = catastrophic damage. Diesel engines are MORE susceptible.

Before water crossing:

- Walk the crossing—check depth, current, bottom
- Maximum safe wading: below air intake (snorkel height)
- Fit water bra/tarp if deep crossing
- Check diff breathers are extended
- Pre-connect recovery strap (ready if stuck)
- Reduce tyre pressure slightly for traction

During crossing:

- Enter slowly, maintain steady speed
- Create bow wave ahead of vehicle
- Never stop mid-crossing
- Keep revs up (exhaust pressure prevents water entry)
- If engine dies: DO NOT try to restart
- Exit slowly—sudden braking = water in brakes

IF ENGINE STALLS IN WATER: Do NOT attempt to restart. You will hydro-lock the engine and destroy it. Get towed out first, then follow the 1HZ recovery procedure below.

⚙️ 1HZ ENGINE HYDRO-LOCK RECOVERY

Specific procedure for your 4.2L 1HZ diesel. Diesel engines suffer MORE damage due to higher compression ratio and torque.

1. **DO NOT CRANK** — Repeated cranking bends connecting rods, the 1HZ's high compression makes this damage severe
2. **Get towed out** — Vehicle must be on dry ground before engine work
3. **Disconnect battery** — Prevents accidental cranking
4. **Remove all 6 glow plugs** — 10mm socket, allows water to escape cylinders
5. **Turn engine BY HAND** — Use 22mm socket on crankshaft pulley bolt, turn clockwise (viewed from front), water will spray from glow plug holes
6. **Repeat several rotations** — Until no more water expelled
7. **Check engine oil** — Pull dipstick: milky white = water contamination = DO NOT START
8. **Check air filter** — If wet, remove and dry or replace
9. **Reinstall glow plugs** — If oil is clean and engine turns freely
10. **Attempt restart** — Only if all checks pass

1HZ-SPECIFIC WARNING: The 1HZ's 22.4:1 compression ratio and heavy rotating mass means hydro-lock damage is more catastrophic than petrol engines. A bent connecting rod is a **minimum R60,000+ repair**. If in doubt, DO NOT START. Tow to a workshop.

Turbo note: Your SAC turbo has its own oil feed. If water entered the intake manifold, the turbo bearings may also be contaminated. Have turbo oil lines inspected before high-load driving.

POST-CROSSING CHECKS

Water causes damage even if you made it through

Immediately after crossing:

- Test brakes gently—apply several times to dry
- Listen for unusual sounds
- Check for water ingress in cabin

At next stop:

- Check diff oils for water contamination (milky colour)
- Check gearbox oil
- Inspect air filter for moisture
- Check fuel filter water trap
- Drain water from anywhere it collected

🌊 FLASH FLOOD RESPONSE

Water levels can rise in minutes

If caught in rising water:

1. Do NOT drive through flooded roads
2. If water rises around vehicle: get out immediately
3. Move to high ground on foot
4. Never cross flowing water on foot if above knee height
5. Abandon vehicle if necessary—your life matters more

30cm of moving water can knock you down. 60cm can float a vehicle. Never underestimate water.

9. COMMUNICATION & EVACUATION

⚖️ WHO OWNS WHAT — READ THIS ONCE, NOW, NOT IN AN EMERGENCY

The same split as fire: R10 owns the drill, R42 owns prevention. Here R10 owns the drill, R29 owns the stack

This document owns the drill. What you do, in what order, what you say, and what you must not do. Everything on this page is written to be executed with a casualty on the ground.

R29 owns the stack. Cover, policies, membership numbers, claims procedure, the four-layer medical model, and the master set of emergency numbers. Nobody reads R29 with a casualty on the ground — that is precisely why it is a separate document.

The numbers printed on this page are a controlled copy. **R29 §6 is the original.** If a number here ever disagrees with R29, R29 wins and this page is wrong.

📞 THE HANDOVER — SAY IT IN THIS ORDER, EVERY TIME

Under stress people give a rambling account and forget the coordinates. Fixed order beats fluency

Twenty seconds, five items, always the same sequence. Read the coordinates off the InReach screen — do not recite them from memory and do not estimate them.

1. **Who is calling.** Your name, and that this is a medical emergency. Give your callback number on the satellite phone before anything else — if the call drops, everything else is recoverable.
2. **What has happened.** One sentence. "Vehicle rollover." "Snakebite." "Suspected cardiac." Not the story of how it happened.
3. **Exact position.** Decimal degrees, read aloud off the InReach, then repeated back and confirmed. Add the nearest airstrip, town or landmark and the surface you are on — AMREF flies fixed-wing and needs a strip.
4. **How many casualties and their condition.** Number of persons. Conscious or not, breathing or not, bleeding controlled or not. Use MIST if you know it; use plain words if you do not.
5. **What you are requesting.** Say the word: evacuation, or medical advice, or both. Do not leave the coordinator to infer it.

Then stop talking and listen. Write down what they tell you — callback interval, aircraft ETA, what to prepare. Keep the InReach text channel open in parallel; it survives conditions that drop a voice call.

⚠️ CALL FIRST. ARGUE COVER LATER.

The disciplined instinct is to verify entitlement before committing to a cost. Here it is the wrong instinct

A coverage question must never delay a clinical decision. Call the evacuation service. Get the aircraft moving. Settle who pays afterwards.

This is written down because the failure mode is a careful person behaving carefully. Faced with an evacuation that might cost more than the vehicle, a prudent operator wants to confirm the membership is current, the zone is covered and the policy will respond — and every minute spent confirming it is a minute the casualty does not have. Entitlement is recoverable after the fact. Physiology is not.

If cover turns out not to respond, that is a debt, and a debt is survivable. Make the call.

The four-layer medical model — who alerts, who flies, who pays, and who carries you on return — is set out in R29 §6. Read it before departure, not now.

SOS SOS ACTIVATION — THE DRILL (MIRRORS R29 §7)

Run both satellite channels simultaneously in a serious emergency — independent networks, independent coordination centres

1 — Activate the InReach SOS first. Hold the SOS button 3 seconds, confirm on screen. Tracking begins immediately and Garmin Response (24/7 IERCC) opens two-way text coordination. Read your GPS coordinates off the InReach screen for the voice call.

2 — Simultaneously call the medevac provider by satellite phone. Voice is faster and clearer than text in a crisis. **East Africa — AMREF Flying Doctors Control Room: +254 730 811 811 / +254 709 962 811 (24/7). Southern Africa — ER24 international dispatch: +27 10 205 3000, aeromedical / flight desk +27 10 205 3588.** *Dial the international forms above from the satellite phone. The SA short-dial codes 082 911 and 084 124 reach the same services from an in-country mobile network but CANNOT be routed by Iridium or Inmarsat — they will fail from a satphone. Numbers owned by R29 §6.* Have ready: names, GPS coordinates (decimal degrees), number of persons, nature of emergency, patient condition and treatment given (MIST), nearest airstrip or landmark, callback number.

3 — Keep both channels live. Reply to Garmin Response texts on the InReach; keep the sat phone on for callbacks. Do not cancel either SOS until rescue is complete or self-rescue is confirmed to BOTH centres.

All numbers, policy references and membership numbers are on the waterproof field card in the glovebox (R29 §9 spec). Full protocol, insurer notification steps and the medevac decision criteria: R29 §7–§8. The Inmarsat IsatPhone 2.1 is owned (purchased 6 Aug 2026, +870 777 751 134). Both the InReach and the satphone are carried on overlanding trips.

📡 COMMUNICATION DEVICES

Options for remote areas where cell coverage is absent

DEVICE	COVERAGE	BEST FOR	LIMITATIONS
Cell Phone	Near towns/roads	Urban/semi-rural areas	No coverage in remote bush
Satellite Phone	Near-global (Inmarsat IsatPhone 2.1 — GEO, owned)	Emergency VOICE calls to AMREF/ER24 — OWNED (purchased 6 Aug 2026; MSISDN +870 777 751 134)	Needs clear sky view toward the equator, antenna extended; no SOS button for us — you dial the provider yourself (R43)

DEVICE	COVERAGE	BEST FOR	LIMITATIONS
HF Radio	Very long range	Overlanding networks, lodges	Requires license, setup. Not carried — undecided as at 4 Oct 2026
PLB/EPIRB	Global (satellite)	Life-threatening emergencies ONLY	One-way SOS only
Satellite Messenger	Global — Iridium LEO (Garmin inReach Mini 2, OWNED)	Primary SOS device · always-on tracking · GPS coordinates · two-way text	Active subscription required — SOS dead if plan suspended (RA-10 verification, R16 §6)

📍 GPS COORDINATES

Always communicate in decimal degrees format

Format for rescue services:

```
-24.123456, 31.654321
```

(Decimal degrees, negative for South/West)

When calling for help, provide (medevac coordinators use MIST — give it in this order):

- Exact GPS coordinates
- Nature of emergency
- MIST: **M**echanism (what happened) · **I**njuries found · **S**igns (conscious? breathing rate? pulse?) · **T**reatment given so far
- Number of persons, any injuries
- Vehicle description and registration
- What assistance you need
- Communication method (phone number, radio freq)

🚁 HELICOPTER LANDING ZONE

If medical evacuation by helicopter

Requirements:

- Flat area minimum 30m × 30m
- Clear of obstacles (trees, wires, vehicles)
- Firm surface (not loose sand)
- Mark centre with X using clothing/material
- Indicate wind direction (smoke, flag, dust)

When helicopter approaches:

- Secure all loose items (will blow away)
- Keep all persons away from LZ
- Shield eyes from rotor wash
- Approach only when signalled by crew
- Approach from front, in pilot's view
- Stay low—never approach from uphill side

✈️ FIXED-WING EVACUATION — the East Africa reality:

AMREF Flying Doctors (the East Africa medevac provider) flies **fixed-wing aircraft, NOT helicopters** — a serviceable airstrip is required, not an open field. The helicopter rules above apply to SA medevac helicopters (ER24) in Southern Africa. For an AMREF evacuation: their Control Room identifies the nearest viable strip from their database — you may need to drive or transport the patient to that strip. Coordinate timing and transport with AMREF Control before moving. Do not waste time clearing a bush LZ for an aircraft that cannot use it (R29 §8, Scenario C).

📞 EMERGENCY CONTACT NUMBERS

Program these before departure—verify they are current

South Africa:

- General Emergency: **112**
- Ambulance: `10177`
- Police: `10111`
- Mountain Rescue: Check local

Botswana:

- Emergency: `999`
- Police: `999`
- Ambulance: `997`

Namibia:

- Emergency: `10111`
- Ambulance: `211111`

Also record: Your travel insurance emergency line · Embassy contact for each country · Medevac providers: AMREF Control Room +254 730 811 811 / +254 709 962 811 (East Africa) · ER24 international: +27 10 205 3000 (Southern Africa) — SA short-dial 082 911 / 084 124 works from an in-country mobile only, never from the satphone · AMREF membership number · Satellite phone numbers for lodges on route — all carried on the R29 waterproof field card

10. FIRST AID KIT SPECIFICATIONS

🩹 OVERLANDING FIRST AID KIT

Beyond standard first aid—designed for remote wilderness travel

This is the trauma / first-aid kit. For the full medical & medication kit — antibiotics, antimalarials & standby treatment, malaria RDTs, dental, allergy and all prescription items — see **R17 (Health & Hygiene) → Medical Kit**. The two lists are complementary, not duplicates.

TRAUMA SUPPLIES:

- Israeli bandages (pressure dressings) ×2

MEDICATIONS:

- Paracetamol (pain/fever)

SPECIALIZED ITEMS:

- Thermometer (digital)

- Tourniquets (CAT or SOFT-T) ×2 — one per person
- Haemostatic gauze (Celox / QuikClot) ×2
- Vented chest seals ×2 (penetrating chest injury)
- Nasopharyngeal airway + lubricant
- Gauze pads (various sizes)
- Gauze rolls (conforming)
- Medical tape (cloth)
- Wound closure strips
- Sterile gloves (multiple pairs)
- Trauma shears
- Tweezers (splinter/tick)
- SAM splint
- Elastic bandages ×2
- Triangular bandage (sling)
- Ibuprofen (pain/inflammation)
- Antihistamines (allergic reactions)
- Oral rehydration salts (ORS)
- Anti-diarrheal (loperamide)
- Antacid
- *Prescription (consult doctor):*
- — Broad-spectrum antibiotic
- — Antimalarials (treatment dose)
- — Adrenaline auto-injector ×2 (carry regardless of known allergy — first anaphylaxis has no history)
- — Personal medications
- Eye wash solution
- Antibiotic eye drops
- Burn gel / hydrogel dressings
- Sunscreen (SPF 50+)
- Insect repellent (DEET)
- Pressure bandage (for snake)
- CPR face shield
- Emergency blanket (space blanket)
- Pen and paper (for notes)
- First aid manual

EXPIRY DATES: Check all medications before each trip. Replace expired items. Keep a checklist with expiry dates in the kit lid.

SNAKE BITE KIT

What you actually need (not venom extractors)

Essential items:

- Pressure bandage (crepe, 10cm wide) ×2
- "Smart bandage" with pressure indicator (ideal)
- Splint material
- Marker pen (to mark bite site, track swelling)
- Notepad (time of bite, symptoms)

NOT NEEDED: Venom extraction devices do not work and waste valuable time. Do not carry them.

Why no antivenom is carried: deliberate. Field antivenom without the ability to run an adrenaline infusion risks killing the patient with anaphylaxis. The pressure bandage buys the hours; the hospital gives the antivenom.

KIT STORAGE & ACCESS

Must be accessible when you need it

- Store in accessible location (not buried under gear)
- Waterproof container or bag
- All persons should know where it is
- Consider two kits: vehicle + day pack
- Include contents list for inventory
- Restock after any use

11. SECURITY INCIDENTS

🚗 CARJACKING RESPONSE

Prioritize your life over the vehicle

IF CONFRONTED:

- **Do NOT resist**—your life is more valuable than the vehicle
- Keep hands visible at all times
- Move slowly, no sudden movements
- Tell them what you're doing: "I'm getting the keys"
- Comply with instructions
- Do not make eye contact (seen as challenge)
- If children in vehicle, say so clearly and calmly

After the incident:

- Get to a safe location
- Contact police immediately
- Document everything you remember
- Contact insurance and embassy

🚧 CHECKPOINT PROCEDURES

Common in Africa—know how to handle

Legitimate checkpoints:

- Stop completely, turn off engine
- Keep hands visible on steering wheel
- Be polite and respectful
- Have documents ready (license, passport, vehicle papers)
- Answer questions calmly and briefly
- Do not offer bribes unprompted

Suspicious checkpoints:

- Trust your instincts
- If possible, do not stop (drive through carefully)
- If you must stop: stay in vehicle, windows up, doors locked
- Be ready to drive away if threatened

🏠 SAFE ZONES

Know where to go in an emergency

Potential safe zones:

- Police stations
- Hospitals / clinics
- Embassies / consulates
- Established lodges and hotels
- Fuel stations (staffed, busy ones)
- Churches / missions

Pre-trip preparation:

- Research security situation for route
- Register with embassy
- Share itinerary with contacts at home
- Know locations of safe zones on route

12. ISOLATION PROTOCOL

⏰ WHEN HELP CANNOT ARRIVE FOR 24-48+ HOURS

Managing extended wait for rescue

Immediate priorities:

1. Ensure vehicle is safe and stable
2. Account for all persons
3. Treat any injuries
4. Inventory all resources (water, food, fuel)
5. Establish communication if possible
6. Create shelter from elements

Resource management:

- **Water:** Ration immediately. 2L/person/day minimum. More if hot.
- **Food:** Less critical than water. Ration but don't skip meals.
- **Fuel:** Conserve for essentials (radio, lights, warmth).
- **Battery:** Turn off non-essential electronics.
- **Morale:** Keep spirits up. Assign tasks. Stay busy.

🏕️ SHELTER CONSIDERATIONS

Protecting from sun, wind, rain, cold

Using the vehicle:

- Vehicle provides shade, wind break, rain protection
- Park to maximize shade (consider sun movement)
- Open windows for ventilation in heat
- Close up at night (animals, insects)

If leaving vehicle area:

- Leave note in vehicle with destination, time, number of persons
- Create shade using tarp, space blanket
- Conserve energy—avoid mid-day sun
- Stay visible for searchers

📢 SIGNALLING FOR RESCUE

Make yourself visible

Visual signals:

- Mirror/reflective surface (signal aircraft)
- Bright clothing laid out on ground
- Fire with green vegetation (smoke)
- SOS ground marker (rocks, branches)
- Vehicle headlights at night

Ground-to-air symbols:

- **V** = Require assistance
- **X** = Require medical help
- **I** = Proceeding in this direction
- → = Proceed in this direction

🧠 MORALE MANAGEMENT

Mental state is critical to survival

- Stay positive—help will come
- Keep busy—assign tasks
- Maintain routine (meals, sleep schedule)
- Take care of each other
- Celebrate small wins
- Focus on what you can control
- Do NOT panic—it wastes energy and impairs judgment

> **Remember:** You have a vehicle, shelter, and resources. Searchers will look for you. Your job is to stay safe and visible until they arrive.

📋 EMERGENCY SUMMARY CHECKLIST

STOP — Think — Observe — Plan
C-ABC: Bleeding, Airway, Breathing, Circulation
Unconscious + breathing: recovery position
Communicate: GPS coordinates, nature of emergency

Stay with vehicle (unless immediate danger)
Ration resources (water, fuel, battery)
Signal for rescue (visual, smoke, mirror)

Snake bite: Pressure bandage for mamba/cobra ONLY
Heat stroke: Cool aggressively, evacuate
Hydro-lock: Do NOT restart engine

CROSS-REFERENCE R26 · Electrical & Wiring | R3 · Spares & Tools | R9 · Vehicle Recovery | R32 · Crank Battery Care